

10_Project_Roadmap_and_Decision_Gates

Kristal Farms Partner Documentation

Document type: Partner decision document

Status: Draft v1

Audience: Hydro/utility partners, infrastructure investors, compute tenants, modular data center suppliers, telecom/fibre partners, community/government partners

Primary geography: Labrador coast

First target: Nain, Nunatsiavut

Expansion logic: Replicate only after Nain validates the coastal hydro/compute/heat model

1. Purpose of this document

This document defines the practical roadmap for moving Kristal Farms from partner alignment to a validated pilot and then to staged expansion across the Labrador coast.

It is written for partners who need to understand:

- what happens first;
- what evidence is required before investment;
- which decisions must be made by partners and the community;
- where technical, community, commercial, and permitting workstreams connect;
- which gates must be cleared before the project advances;
- how the project avoids premature commitments, overbuilding, or unvalidated claims.

The roadmap is not a construction schedule. It is a **decision framework**. Its purpose is to prevent the project from moving faster than evidence, consent, engineering, fibre, heat offtake, or economics allow.

2. Roadmap principle

Kristal Farms should advance only through gated evidence.

The project thesis is clear: place modular compute near Labrador coastal heat users, connect to local hydro where feasible, export compute by fibre, recycle server heat locally, and expand by pads only when the technical and community case is proven.

The roadmap therefore follows five rules:

1. **Nain first**

Nain is the first target because it best tests the full Labrador coast model: local/coastal hydro opportunity, village heat demand, marine logistics, fibre dependency, community process, and modular compute deployment.

2. **Community process before project lock-in**

Site confirmation, layout, heat allocation, governance, and benefit-sharing must remain open until community and partner processes define them.

3. **Heat demand gates compute expansion**

Compute pads should not expand faster than confirmed useful heat sinks, cooling/rejection limits, fibre service, and power capacity.

4. **Pilot before replication**

Labrador coastal replication should begin only after Nain validates the model against measurable operating metrics.

5. **Reversibility by design**

The project should use pads, containers, modular heat equipment, metered interfaces, and staged deployment so that the system can be expanded, paused, reconfigured, or removed without permanent overbuild.

3. Roadmap at a glance

Phase	Name	Main question	Primary output
Phase 0	Partner alignment	Is there a credible coalition to investigate the project?	Partner alignment memo

Phase	Name	Main question	Primary output
Phase 1	Site confirmation	Can Nain support a technically credible first project?	Nain site confirmation pack
Phase 2	Community / FPIC process	Is there a legitimate path to community consent and benefit design?	Community process plan and consent roadmap
Phase 3	Hydro and grid pre-feasibility	Can power be supplied without a long HV megaproject?	Hydro/grid pre-feasibility report
Phase 4	Fibre and logistics design	Can compute containers be connected, delivered, serviced, and repaired?	Fibre/logistics feasibility report
Phase 5	Pilot pad	Can one pilot pad operate safely and measurably?	Commissioned pilot pad and operating report
Phase 6	Heat loop and greenhouse	Can server heat deliver visible community value?	Heat delivery and greenhouse integration report
Phase 7	Expansion pads	Can the project expand by evidence-gated pads?	Expansion investment package

4. Decision gates at a glance

Gate	Name	Cleared when	Decision
Gate A	Partner interest	Partner roles, data access, and diligence scope are agreed	Proceed to structured feasibility
Gate B	Site data validated	Nain site, hydro, heat users, land, fibre, and logistics have enough evidence for pre-feasibility	Proceed to community/process and technical studies
Gate C	Community process initiated	FPIC/community pathway, governance topics, and benefit principles are formally opened	Proceed only within agreed community process
Gate D	Pilot economics approved	Pilot-scale CAPEX/OPEX, revenue basis, heat value, risks, and partner commitments are credible	Proceed to pilot readiness
Gate E	Construction readiness	Permits, land, fibre, power, heat loop, procurement, EPC/O&M roles, and financing are ready	Proceed to construction or defer

A gate is not a public-relations milestone. It is a decision point where the project either advances, pauses, redesigns, or stops.

5. Phase 0 — Partner alignment

Objective

Establish whether the right partners are at the table to evaluate a Nain-first Labrador coast project.

Core work

- Confirm the project thesis: Labrador coast, Nain first, coastal replication later.
- Identify the partner type being approached:
 - hydro/utility partner;
 - infrastructure investor;
 - compute tenant;
 - modular data center supplier;
 - telecom/fibre partner;
 - community/government partner.
- Define what Kristal Farms brings.
- Define what each partner brings.
- Confirm the first diligence scope.
- Define confidentiality and data-room access level.
- Separate external-facing claims from internal assumptions.

Required evidence

- Partner contact list and roles.
- Initial data-request list.
- Draft NDA or confidentiality approach if needed.
- Source document index.
- Initial issue list: hydro, fibre, heat, land, logistics, governance, permitting, commercial model.

Deliverables

- Partner alignment memo.
- Initial diligence request list.
- Partner role matrix.

- Meeting decision record.

Gate A — Partner interest

Proceed if:

- at least one credible partner is willing to engage in structured diligence;
- the partner understands that this is a Labrador coastal project, not a generic national AI data center proposal;
- the partner accepts that Nain is the first target and that later replication is conditional;
- the partner agrees that community process and technical validation come before investment lock-in.

Do not proceed if:

- the partner wants a speculative national/global package before the Labrador/Nain case is validated;
 - the partner requires unvalidated financial claims;
 - the partner wants to bypass community process;
 - the partner frames the project as a Churchill Falls/inland megaproject strategy.
-

6. Phase 1 — Site confirmation

Objective

Confirm whether Nain has enough evidence to justify full pre-feasibility work.

Core work

- Confirm the candidate hydro basis and whether the Fraser River/Nain concept remains the right first hydro-screening reference.
- Map the relationship between hydro source, village, port, heat users, potential pad yard, and fibre path.
- Identify candidate land areas for:
 - village substation;
 - compute pad yard;
 - heat exchanger station;
 - thermal storage;

- public-building heat connections;
- greenhouse site;
- port staging/logistics.
- Identify constraints:
 - seasonal access;
 - shoreline and port limitations;
 - environmental sensitivities;
 - existing utility infrastructure;
 - known community priorities;
 - fibre constraints;
 - emergency backup and diesel dependency.

Required evidence

- Site inventory.
- Hydro candidate summary.
- Heat-user inventory.
- Initial load and heat demand screen.
- Fibre availability screen.
- Logistics and sealift screen.
- Land and permitting screen.
- Environmental and cultural sensitivity screen.

Deliverables

- Nain site confirmation pack.
- Site risk memo.
- Preliminary heat-user map.
- Preliminary pad-yard and substation siting options.
- Diligence gap list.

Gate B — Site data validated

Proceed if:

- Nain remains a credible first site after basic screening;
- the hydro, heat, fibre, land, and logistics questions are defined well enough for pre-feasibility;
- the project can still be designed around village heat users rather than a remote dam-only site;
- major red flags are known and can be studied rather than ignored.

Do not proceed if:

- the only feasible compute location is too far from heat users;
 - fibre has no credible path to acceptable service;
 - hydro/site assumptions are too weak to support pre-feasibility;
 - community process cannot begin legitimately;
 - the project would require a long HV transmission strategy inconsistent with the thesis.
-

7. Phase 2 — Community / FPIC process

Objective

Start a legitimate community process before technical and commercial decisions become fixed.

Core work

- Present the concept in plain language:
 - what is being proposed;
 - what is not being proposed;
 - where the project could be located;
 - what heat reuse means;
 - what the host can see and cannot see under black-box tenancy;
 - what decisions remain open.
- Define the FPIC/community engagement pathway.
- Identify decision-makers and community bodies.
- Identify required local studies, meetings, translations, and consultation formats.
- Define draft benefit categories:
 - heat allocation;
 - jobs and training;
 - connectivity improvements;
 - local procurement;
 - community revenue or benefits;
 - greenhouse/food benefit;
 - environmental monitoring;
 - grievance process.
- Identify governance questions that are not yet final.

Required evidence

- Community engagement plan.
- FPIC process roadmap.
- Plain-language project summary.
- Open design item register.
- Initial community benefits framework.
- Grievance and dispute-process options.

Deliverables

- Community process plan.
- FPIC/consent roadmap.
- Draft community benefits principles.
- Governance design issue list.
- Public-facing FAQ.

Gate C — Community process initiated**Proceed if:**

- the community process is formally opened through legitimate channels;
- the project is described honestly as preliminary and evidence-gated;
- community benefits are discussed before final commercial structuring;
- unresolved governance items are labelled as open, not finalized;
- there is a path for objections, modifications, or refusal.

Do not proceed if:

- community process is treated as a formality;
 - heat allocation is decided without community input;
 - governance councils or committees are presented as finalized when they are not;
 - the project cannot explain black-box tenancy, data boundaries, or infrastructure monitoring clearly.
-

8. Phase 3 — Hydro and grid pre-feasibility

Objective

Determine whether local hydro and village electrical integration can support a staged compute project without becoming a long-transmission megaproject.

Core work

- Validate hydrology, seasonal flow, and preliminary generation potential.
- Evaluate run-of-river or other feasible configuration options.
- Identify environmental constraints and permitting pathway.
- Evaluate the short MV feeder concept from hydro source to village substation.
- Define substation location and preliminary one-line architecture.
- Establish load sequencing and pad energization strategy.
- Reserve power for local/community needs before compute expansion.
- Confirm backup concept for critical loads.
- Define metering boundaries:
 - hydro output;
 - substation input/output;
 - pad consumption;
 - heat equipment load;
 - auxiliary loads.

Required evidence

- Hydrology and flow data requirements.
- Conceptual hydro layout.
- Preliminary electrical single-line diagram.
- Load and power quality requirements.
- Interconnection concept.
- Local reserve margin concept.
- Diesel backup / emergency plan.
- Permitting pathway memo.

Deliverables

- Hydro/grid pre-feasibility report.
- Power interconnection concept.
- Local load reserve policy draft.
- Preliminary metering plan.

- Environmental permitting issue list.

Advancement condition

The project should advance only if it can preserve the core design: **local power use, short electrical connection where feasible, village heat value, and staged compute expansion.**

9. Phase 4 — Fibre and logistics design

Objective

Confirm whether the project can reliably export compute results and physically deploy/maintain modular containers on the Labrador coast.

Core work

Fibre

- Identify existing fibre or high-capacity connectivity options.
- Confirm latency, uptime, service-level constraints, and repair model.
- Identify redundancy options.
- Define NOC requirements.
- Define tenant handoff and black-box monitoring boundary.
- Confirm what infrastructure metrics can be monitored without tenant-data exposure.

Logistics

- Confirm marine delivery route and seasonal limitations.
- Identify port/quay handling constraints.
- Define container dimensions, weights, lifts, and staging.
- Define winterization and storage requirements.
- Identify repair, replacement, and spare-parts strategy.
- Confirm construction window assumptions.

Required evidence

- Fibre feasibility memo.
- NOC and connectivity concept.
- Telecom partner options.
- Sealift/logistics assessment.

- Port staging plan.
- Container delivery and replacement plan.
- Seasonal access risk register.

Deliverables

- Fibre/logistics feasibility report.
- Connectivity SLA concept.
- Marine logistics plan.
- Container deployment method statement.
- Operations support plan.

Advancement condition

The project should not treat compute demand as bankable until fibre and logistics are credible. Power alone is not enough; the product is compute delivered over reliable connectivity.

10. Phase 5 — Pilot pad

Objective

Commission the smallest credible operating unit that proves the integrated system without overbuilding.

Pilot scope

The pilot should include:

- one modular compute pad or equivalent test load;
- metered power handoff;
- cooling interface;
- black-box monitoring boundary;
- basic NOC visibility;
- controlled heat capture;
- safe rejection path for surplus heat;
- public dashboard inputs;
- incident and curtailment procedures;
- reversibility plan.

The pilot may use a real tenant, test compute load, or staged load bank depending on partner readiness and data/security requirements.

Required evidence

- Pilot CAPEX/OPEX estimate.
- Pilot revenue or tenant-demand basis.
- Pilot energy and heat balance.
- Pilot risk register.
- Construction readiness checklist.
- Operations staffing plan.
- Commissioning plan.
- Measurement and verification plan.
- Cyber/tenant boundary plan.

Deliverables

- Pilot investment memo.
- Pilot design package.
- Pilot commissioning plan.
- Pilot operating procedures.
- Pilot measurement and verification framework.
- End-of-pilot report template.

Gate D — Pilot economics approved

Proceed if:

- pilot cost, funding, partner roles, and risk allocation are credible;
- the pilot has a clear reason to exist beyond publicity;
- the pilot can measure power, heat, uptime, fibre, operations, and community value;
- the pilot can be stopped or removed if it does not meet criteria;
- heat-first operation is built into procedures and contracts.

Do not proceed if:

- the pilot requires unvalidated full-scale economics;
- the tenant model is unclear;
- the heat path is symbolic rather than operational;
- fibre service cannot support the intended compute use;
- community process has not advanced enough to support deployment.

11. Phase 6 — Heat loop and greenhouse

Objective

Convert server heat from a technical by-product into measurable community value.

Core work

- Connect first-priority public buildings or heat users.
- Define heat allocation priorities.
- Install or validate building substations, heat exchangers, meters, and controls.
- Integrate thermal storage where useful.
- Define domestic hot water and radiator requirements.
- Confirm whether booster heat pumps are required for legacy systems.
- Define greenhouse role as seasonal heat sink and food-production asset.
- Define summer surplus strategy: reuse, store, then reject.
- Confirm environmental constraints for any heat rejection.

Required evidence

- Heat-user inventory.
- Building connection plan.
- Heat-loop hydraulic concept.
- Heat-metering plan.
- Greenhouse concept.
- Thermal storage concept.
- Diesel-displacement calculation method.
- Heat Utilization Factor methodology.

Deliverables

- Heat loop implementation report.
- Greenhouse integration plan.
- Heat allocation policy draft.
- MWh_{th} and diesel-avoided reporting method.
- Summer surplus operating procedure.

Advancement condition

The project should expand compute only when useful heat sinks, storage, or permitted rejection pathways can support the thermal load. Heat reuse is not optional branding; it is a gating part of the model.

12. Phase 7 — Expansion pads

Objective

Expand from pilot to multiple pads only after operating evidence supports the next increment.

Core work

- Review pilot metrics against thresholds.
- Confirm additional power capacity.
- Confirm additional heat sink capacity.
- Confirm fibre headroom and redundancy.
- Confirm tenant demand or committed occupancy.
- Confirm operational staffing and training.
- Confirm spare-parts and maintenance model.
- Confirm environmental monitoring results.
- Confirm governance and benefits reporting.
- Update commercial model and risk register.

Required evidence

- Pilot operating report.
- Expansion power/heat/fibre balance.
- Tenant or demand evidence.
- Updated CAPEX/OPEX.
- Updated community benefits plan.
- Updated risk register.
- Updated permits and construction plan.

Deliverables

- Expansion investment package.
- Pad expansion plan.

- Updated SLA and tenant framework.
- Updated monitoring and public dashboard plan.
- Replication screening criteria for next Labrador coastal community.

Expansion rule

Expansion should occur **by pads**, not by speculative overbuild.

Each additional pad should pass four tests:

1. **Power test** — power is available without compromising local needs.
2. **Heat test** — heat can be used, stored, or rejected within limits.
3. **Fibre test** — connectivity can support the compute service.
4. **Consent/value test** — the community benefit case remains legitimate and visible.

13. Gate E — Construction readiness

Gate E applies before major construction, pilot buildout, or any expansion stage.

Gate E checklist

Workstream	Must be ready
Community	FPIC/community process status is documented; open items are not hidden
Land	Site access, tenure, and layout are clear enough for construction
Hydro/power	Power source, interconnection, protection, metering, and reserve margin are defined
Fibre	Connectivity provider, service path, uptime target, repair model, and NOC interface are defined
Heat	First heat users, heat loop, meters, storage/rejection path, and operating hierarchy are defined

Workstream	Must be ready
Environment	Permitting path, baseline studies, and monitoring requirements are defined
Logistics	Sealift, port staging, lifts, storage, and winterization are planned
Commercial	Partner roles, funding, revenue basis, SLA structure, and risk allocation are documented
Operations	Staffing, training, maintenance, incident response, and reporting are ready
Reversibility	End-of-lease and removal/repurposing plan is documented

Gate E decision

At Gate E, the project should make one of four decisions:

1. **Proceed** – construction readiness is adequate.
2. **Proceed with conditions** – specific evidence or approvals must be closed before mobilization.
3. **Redesign** – core assumptions have changed and the design must be revised.
4. **Defer or stop** – the project is not ready or is no longer justified.

14. Workstream responsibility matrix

Workstream	Lead partner type	Supporting partners	Key output
Project coordination	Kristal Farms / project sponsor	All partners	Integrated roadmap and gate records
Community process	Community/government partner	Kristal Farms, legal, technical advisers	FPIC/community process roadmap

Workstream	Lead partner type	Supporting partners	Key output
Hydro resource	Hydro/utility partner	Engineers, environmental advisers	Hydro pre-feasibility
Electrical integration	Hydro/utility partner	EPC, pad supplier	MV/substation/inte rconnection concept
Compute pads	Modular data center supplier	Tenant, EPC, Kristal Farms	Pilot pad design and deployment
Fibre/NOC	Telecom/fibre partner	Tenant, Kristal Farms	Connectivity and NOC plan
Heat loop	Heat/engineering partner	Community, building owners, greenhouse partner	Heat reuse implementation plan
Commercial model	Infrastructure investor / Kristal Farms	Hydro, tenant, community partners	Pilot and expansion economics
Risk/permitting	Project sponsor / legal	Technical and community partners	Risk register and permit path
Metrics/reporting	Kristal Farms / governance bodies	Operators, community representatives	Public dashboard and reports

15. Evidence package by gate

Evidence package	Gate A	Gate B	Gate C	Gate D	Gate E
Partner role matrix	Required	Update	Update	Update	Finalize
Source/data-room index	Required	Update	Update	Update	Finalize
Nain site screen	Initial	Required	Update	Update	Finalize for pilot
Hydro/grid screen	Initial	Required	Inform	Required	Finalize for construction
Fibre/logistics screen	Initial	Required	Inform	Required	Finalize for construction
Community process plan	Initial	Draft	Required	Required	Status confirmed
Heat-user inventory	Initial	Required	Update	Required	Finalize first connections
Pilot economics	Not required	Initial	Inform	Required	Final construction version
Risk register	Initial	Required	Required	Required	Construction version
Permitting path	Not required	Initial	Inform	Required	Mobilization-ready
Measurement plan	Not required	Initial	Inform	Required	Commissioning-ready

16. Metrics used to judge progress

The roadmap should be judged by a stable set of metrics, not by narrative momentum.

Technical metrics

- PUE
- WUE
- pad uptime
- power availability
- fibre availability
- latency p95
- cooling loop ΔT and flow
- heat-loop availability
- incident count and response time

Heat and community metrics

- MWh_{th} delivered
- Heat Utilization Factor
- diesel avoided
- public buildings connected
- greenhouse operating days
- greenhouse output
- local jobs
- training hours
- local procurement
- community benefit commitments delivered

Commercial metrics

- pad occupancy
- tenant commitments
- operating cost per pad
- maintenance cost trend
- expansion CAPEX per pad
- avoided fuel value
- heat service value

Governance metrics

- community meetings completed

- open design items resolved
 - grievances received/resolved
 - public dashboard updates
 - quarterly reviews
 - annual reporting
-

17. Open decisions to resolve during the roadmap

The following items should not be presented as finalized until the relevant phase closes them:

1. Final Nain site layout.
 2. Final hydro configuration.
 3. Final fibre path and redundancy model.
 4. Final pad count and expansion capacity.
 5. Final heat-user list and heat allocation priorities.
 6. Final greenhouse role and operator.
 7. Final governance committee structure and powers.
 8. Final commercial partnership structure.
 9. Final tenant eligibility and SLA terms.
 10. Final environmental permitting conditions.
 11. Final community benefits agreement / IBA structure.
 12. Final replication sequence after Nain.
-

18. Conditions that should stop or pause the project

The roadmap should protect the project from advancing under weak assumptions.

The project should pause, redesign, or stop if:

- community consent cannot be pursued through a legitimate process;
- hydro potential cannot support even a credible staged pilot;
- the project requires long HV transmission inconsistent with the coastal thesis;
- fibre service cannot support the intended compute use;
- compute containers must be located too far from heat users;
- heat rejection becomes the primary operating mode rather than a backup path;

- permitting risks become disproportionate to pilot scale;
 - pilot economics require speculative full-scale assumptions;
 - partners require claims that cannot be validated;
 - governance and benefit-sharing cannot be made credible.
-

19. Replication after Nain

Replication should be treated as a second-stage strategy, not a promise.

After the Nain pilot produces evidence, Kristal Farms can screen other Labrador coastal communities such as Hopedale, Makkovik, Postville, and Rigolet using the same criteria:

- hydro or renewable power potential;
- village heat demand;
- coastal/marine access;
- fibre path;
- land and port staging;
- community process;
- environmental constraints;
- pad expansion potential;
- reversibility;
- measurable community value.

The replication question is not “where else can we put containers?”

The correct question is:

Which coastal community can reproduce the full model: power, heat, fibre, logistics, consent, and measurable local benefit?

20. Recommended immediate next steps

The next practical actions are:

1. Confirm the first partner audience for the roadmap.

2. Build the Nain data-request list.
 3. Prepare the partner role matrix.
 4. Prepare a one-page Gate A decision memo.
 5. Open the site confirmation workstream for Nain.
 6. Prepare a community-facing plain-language concept note.
 7. Build the initial hydro/fibre/heat/logistics risk register.
 8. Identify which documents belong in the first data room.
-

21. Source basis

This roadmap is based on the current Kristal Farms source corpus, especially:

- **Kristal Farms Internal Reference Document** — technical architecture, phasing, operations, metrics, monitoring, risk management, and village-first infrastructure model.
 - **Documentation Kristal Farms** — black-box tenancy, SLA principles, governance questions, FPIC/community process, metrics, and unresolved design items.
 - **Kristal Farms — Heat Recycling Plan** — heat-first implementation, reuse → store → reject hierarchy, village-first siting, two sealed circuits, building heat, greenhouse, thermal storage, and metrics.
 - **Kristal Farms — Cost Advantage & Strategic Rationale** — cost logic, short MV feed, natural cold, heat value, marine logistics, modular pads, and export compute rather than electricity.
 - **Potentiel hydroélectrique isolé au Nunavik et au Labrador** — Labrador/Nain hydro-screening basis and exclusion logic for large/inland/non-coastal hydro distractions.
 - **EC8ADFDE PDF synthesis** — partner-facing framing for modular cold-climate hydro compute, black-box tenancy, FPIC, metering, heat reuse, and phased deployment.
-

22. Partner decision ask

Partners are not being asked to approve full construction at this stage.

The immediate ask is narrower:

Approve a structured Gate A → Gate B diligence process for Nain, with clear partner roles, source-data access, community-process preparation, and evidence requirements for whether the project should advance.

If Gate B and Gate C are not cleared, the project should not proceed to pilot economics.

If Gate D and Gate E are not cleared, the project should not proceed to construction.